








IPESE
Industrial Process and Energy Systems Engineering

# Process Integration for Industry Decarbonization: Enabling a Shared Database of Ex-Ante Models

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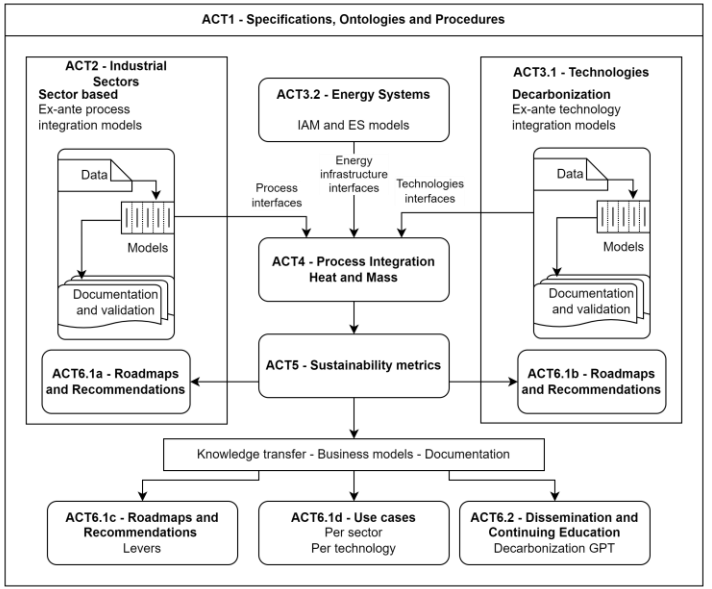
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# Process (and Energy Systems!) Integration for industry decarbonization

## Subtask 1

**Enabling a shared database of ex-ante models of industrial processes, decarbonization technologies and energy systems featuring sustainability metrics**

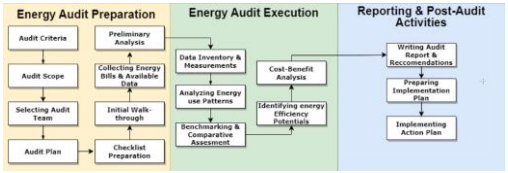
1. Develop **ontologies, specs and procedures** to generate, validate, and share models.
2. Share an **open-source database** of ex-ante models of key **industrial sectors.**
3. Share **energy technology** models to promote renewable energy use in industry.
4. Share **energy systems** models, distribution networks and seasonal demand data.
5. Document process **integration methods/tools** for weighing **decarbonization pathways**
6. Build **Reporting and documentation*** tool of integration results to outline roadmaps and symbiosis.
7. Share experiences through **workshops**, scientific events, and **educational programs**.




Metadata, data, models, references, visualization, hosting, maintainability, confidentiality, scalability*

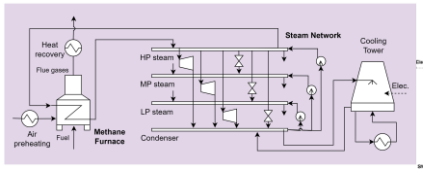
# Database generation

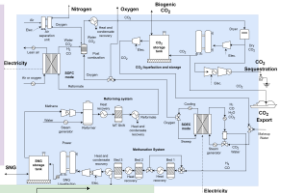
## Industrial process DB

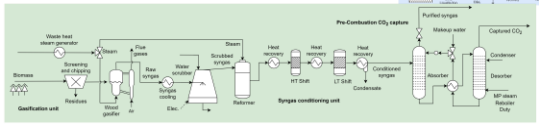



- Collection, benchmark and manage data: flowsheeting, industry validation & constraints

## Utility systems and decarbonization technologies DB



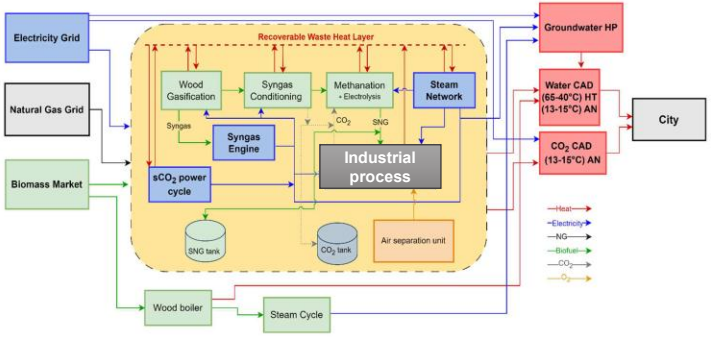





- Inventory and detailed description of process unit operations (process needs and interfaces)

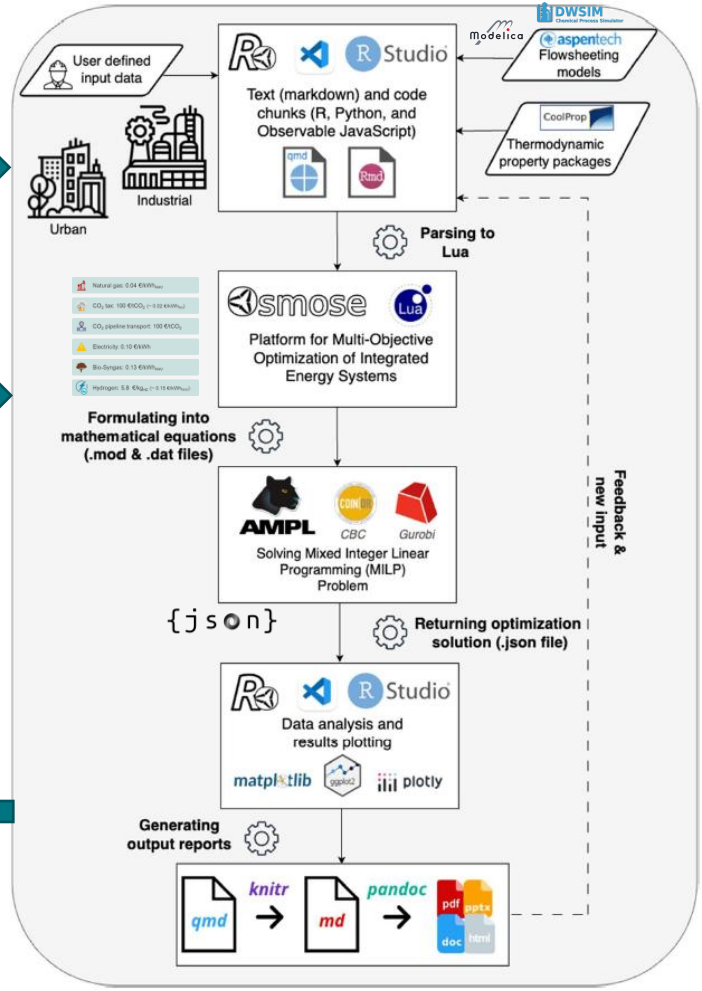
- Characterization of mass, energy, information, emissions, costs flows.

## Integrated processs, technologies and energy systems




- Propose energy efficiency upgrading and decarbonization options

- Issue recommendations on decarbonization roadmaps



# Models database



AIDRES METHAREN
EPOS ANNEX IEA
MASBOOK NETZEROLAB
TRILATE SCCER



- Metadata: Background, Author, Version, Confidentiality.
- Context: Connections, Flows, Quality, Cost, Impact, TRL.
- Models: Parameters, decision variables and assumptions.
- Metrics: Energy, environmental, economic indicators

- **Aluminium process**
- **Sugar process**
- **Waste upgrading**
- **Olefins process**
- **Watchmaking process**
- Dairy process
- Brewery process
- Explosives process
- Malting process
- Tannery process
- Distillery process
- Fertilizers process
- Cement process
- Steel process
- Glass process
- Refinery process
- Lime process
- Biogas production
- Fischer Tropsch
- Methanol production
- Gasification
- Reforming and rWGS
- $CO_2$ seq. and injection
- Carbon capture
- Electrolyzers and fuel cells
- Heat pumping (MVR,HPs)
- PV panels
- Wind turbines
- Air separation
- Methanation
- Oxyfuel technology
- Conventional heaters
- Internal comb. engines
- Gas turbines
- Steam/ORCs
- Cogeneration
- Steam network
- Cooling systems
- Air separation
- Refrigerator, etc.

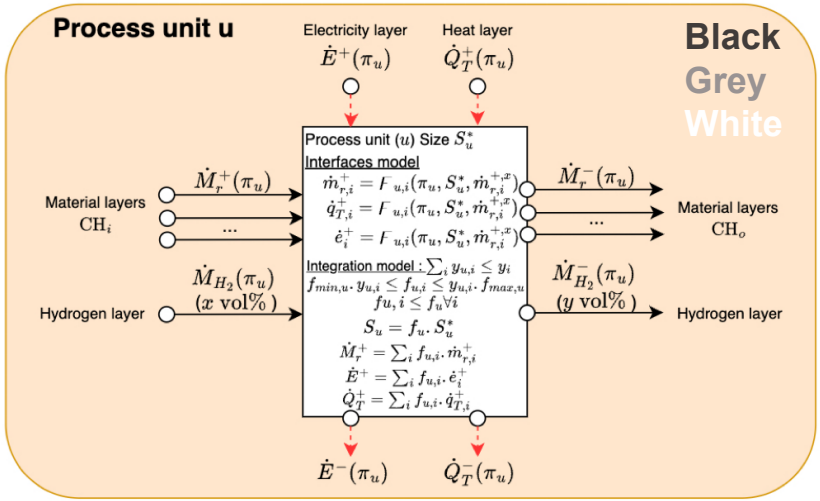
**System integration model**

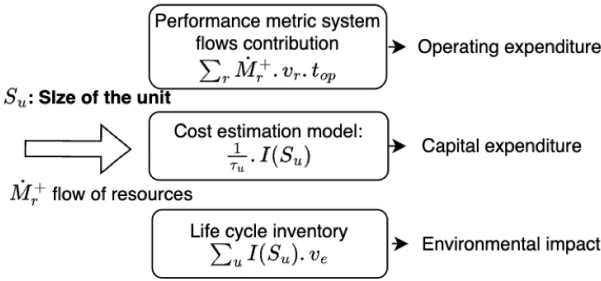
Mass balance: $\sum_u \dot{M}_r^+(\pi_u) = 0$

Energy balance: $\sum_u \dot{E}^+(\pi_u) = 0$

Heat cascade: $\sum_u \dot{Q}_T^+(\pi_u) + R_T - R_{T-1} = 0$

$S_u^*$ : Nominal size of unit u
$y_{u,i}$ : existence of interface i of unit u
$f_{u,i}$ : level of use of interface i of unit u
$f_u$ : level of use of unit u
$\pi_u$ : parameters of interface model of unit u
$= \pi_u^D$ : Decision variable
$= \pi_u^K$ : from Knowledge
$= \pi_u^M$ : fitted from Measurement
$= \pi_u^C$ : from Context



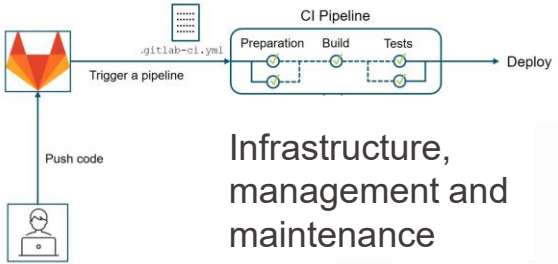



# Deployment and maintenance

Syntax, terminology and specifications



Infrastructure, management and maintenance




```
# Heat Pump {-}

```{rosmose heatpump-et}
! OSMOSE ET heatpump
```

```{rosmose}
Evap_Tin = 55.5 [C] # Evaporator temperature inlet
Evap_Tout = 55.5 [C] # Evaporator temperature outlet
Cond_Tin = 74.5 [C] # Condenser temperature inlet
Cond_Tout = 74.5 [C] # Condenser temperature outlet
Cond_Qmax = 6000 [kW] # Condenser reference heat flow rate
exeff = 0.5 [-] # Second law efficiency
dtmin_2ph = 2 [C] # phase-change delta t minimum
n = 40.0 [yr] #lifetime of a heat pump
i = 0.06 [-] #interest rate
CEPCI_2020 = 596.2 [-] # actual CEPCI
CEPCI_2008 = 575.4 [-] # CEPCI 2008
```

```{rosmose}
COPcarnot = (%Cond_Tin% + 273) / (%Cond_Tin% - %Evap_Tin%) [-] # Carnot COP
COP = %exeff% * %COPcarnot% [-] # Actual COP
W_heatpump = %Cond_Qmax% / %COP% [kW] # Heat pump power consumption
Evap_Qmax = %Cond_Qmax% * (%COP% - 1) / %COP% [kW] # Evaporator heat flow rate
Annuity = (%i%*(1+%i%)**%n%)/((1+%i%)**%n%-1) [-] #annualization factor
Cinv2_HP = 300*%Cond_Qmax%*(%CEPCI_2020%/%CEPCI_2008%)*%Annuity% [Euro/y] #300
```

*Resource Streams*
Defining the resource streams, in this case electricity to

```{rosmose HeatPump_rs}
: OSMOSE RESOURCE_STREAMS HeatPump

|layer      |direction|value               |
|:----------|:--------|:-------------------|
|ELEC       | in      |%W_heatpump%        |
```

*Heat Streams*

```{rosmose HeatPump_hs}
: OSMOSE HEAT_STREAMS HeatPump

|name          |Tin        |Tout       |Hin        |Hout       |DT min/2   |alpha|
|:-------------|:----------|:----------|:----------|:----------|:----------|:----|
|evaporation   |%Evap_Tin% |%Evap_Tout%|0          |%Evap_Qmax%|%dtmin_2ph%|1    |
|condensation  |%Cond_Tin% |%Cond_Tout%|%Cond_Qmax%|0          |%dtmin_2ph%|1    |
```

## Visualization of Heat Pump {-}
```

- Introduction
- Problem Definition
- Dairy Process
- Cooling Tower
- **Heat Pump**
  - Visualization of Heat Pump
- Heat Pump 2
- Heat Pump 3
- Heat Pump 4
- Market
- Optimization Results

## Heat Pump

Note: Heat pumping will be profitable only if it allows to transfer heat from below to above the pinch, i.e., if it transforms excess heat from the system heat source into useful heat for the system heat sink.

**Layers of the Heat Pump ET**

OSMOSE LAYERS heatpump

| Layer | Display name | shortname | Unit | Color |
|---|---|---|---|---|
| ELEC | Electricity | elec | kW | yellow |

**Heat Pump unit of the Heat pump ET**

OSMOSE UNIT heatpump

| unit name | type |
|---|---|
| HeatPump | Utility |

**Parameters of the Heat Pump unit**

| cost1 | cost2 | cinv1 | cinv2 | imp1 | imp2 | fmin | fmax |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 123955.26918427552 | 0 | 0 | 0 | 10 |

**Heat pump Streams**

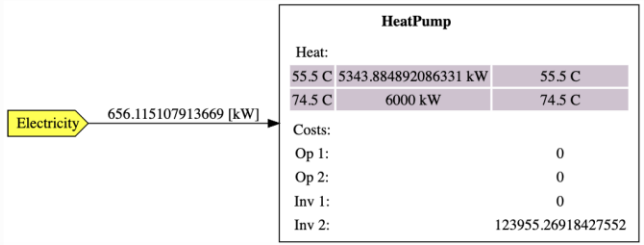
*Resource Streams* Defining the resource streams, in this case electricity to the heatpump

| layer | direction | value |
|---|---|---|
| ELEC | in | 656.115107913669 |

*Heat Streams*

| name | Tin | Tout | Hin | Hout | DT min/2 | alpha |
|---|---|---|---|---|---|---|
| evaporation | 55.5 | 55.5 | 0 | 5343.884892086331 | 2 | 1 |
| condensation | 74.5 | 74.5 | 6000 | 0 | 2 | 1 |

## Visualization of Heat Pump




heatpump-visualize

# Selected industrial sectors



- Heavy industry sector: Secondary aluminium



- Chemicals sector: Olefins production



- Energy sector: Biowaste valorization



- Food sector: Sugar production



- Luxury sector: Watchmaking



# Aluminium sector

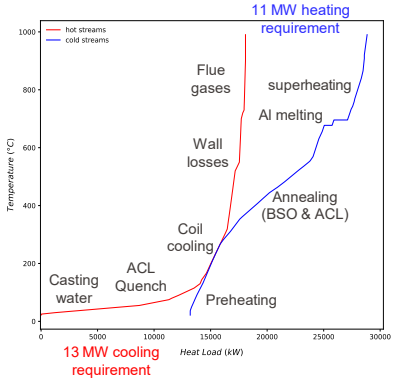
245 $kt_{AlBSO}/y$

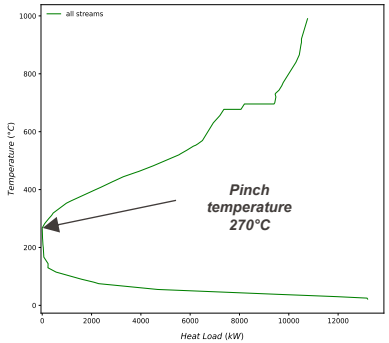
193 GWh/y of natural gas

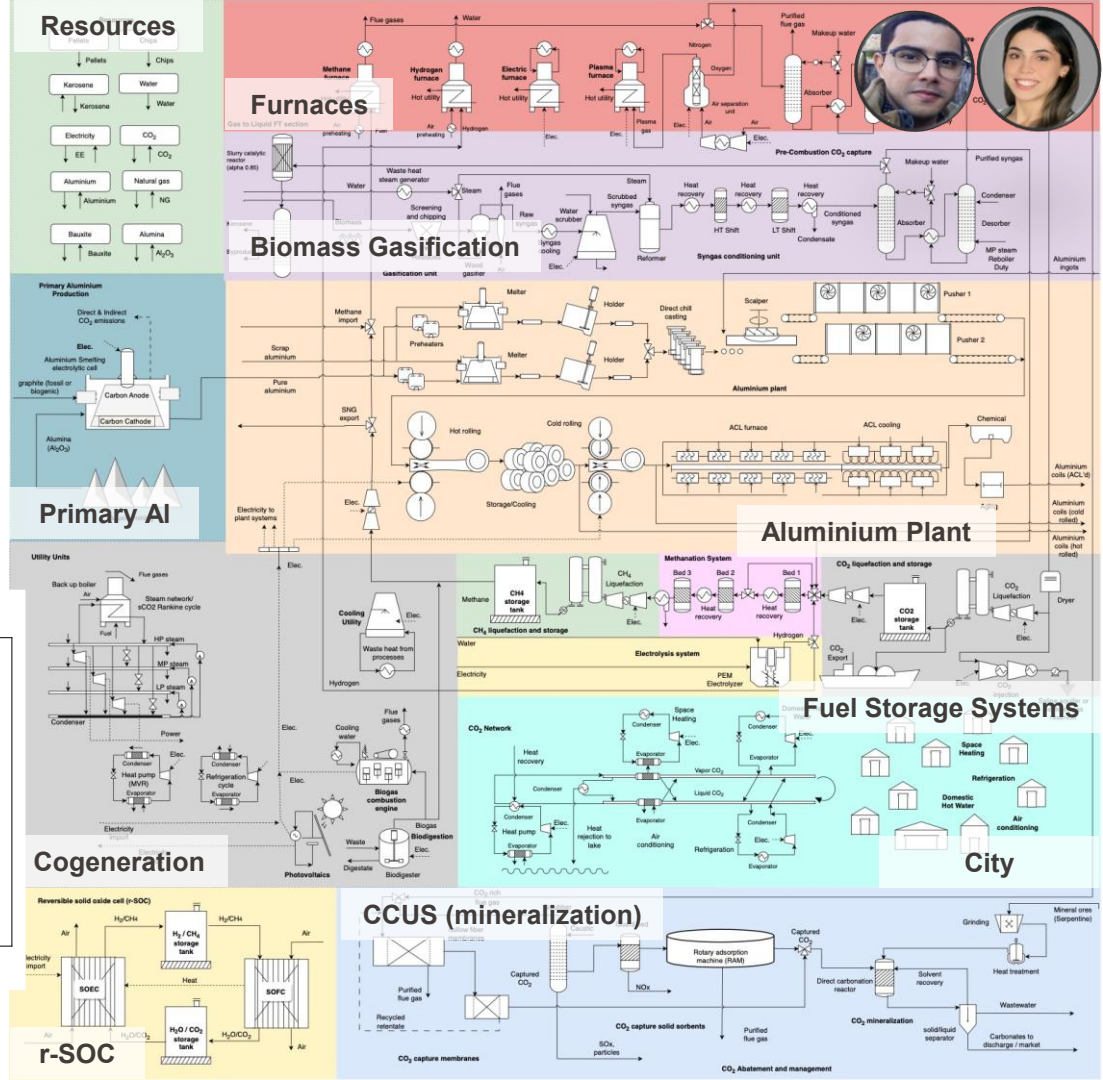
70 GWh/y of electricity

50.3 kt $CO_2$/y

- Utility options
- Achieve net-zero targets
- Systemic approach

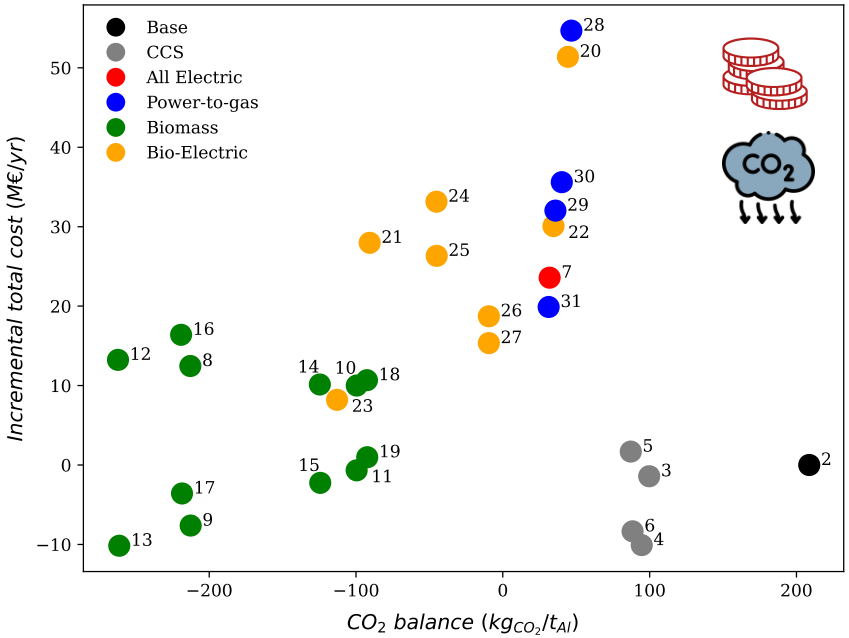


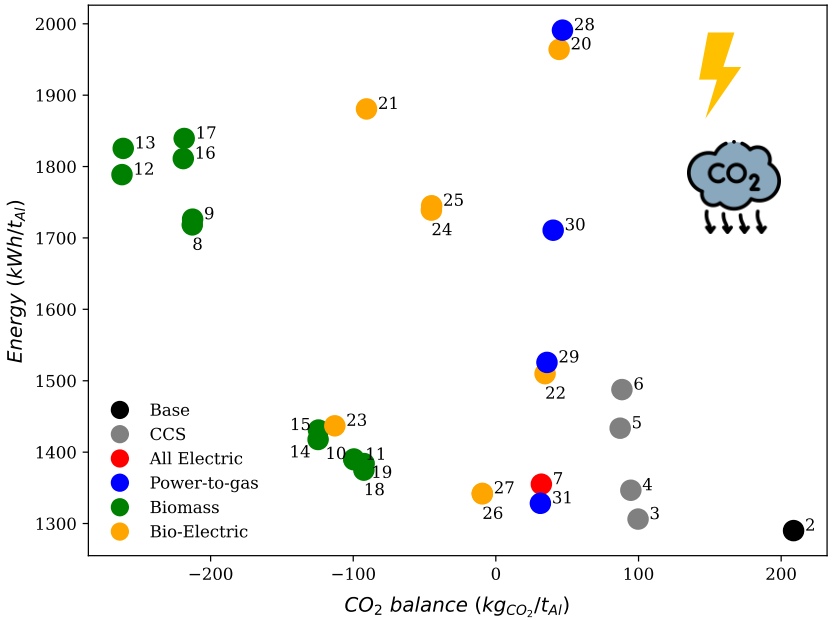



# Aluminium sector

*30+ decarbonization configurations*




**Economic performance vs. Net emissions**



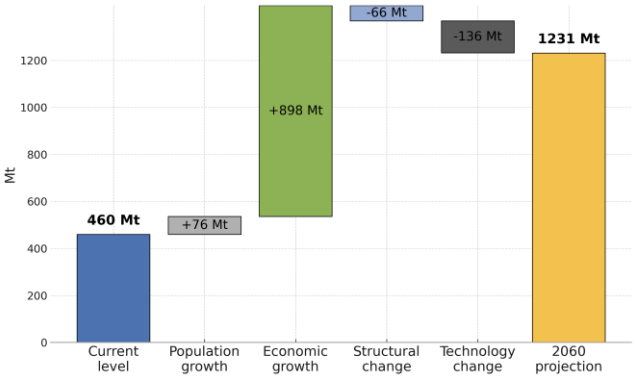

**Energy Performance vs. Net emissions**

*100 €/t$CO_2$ tax, 0.15 €/k$Wh_{EE}$, 0.045 €/k$Wh_{NG}$, 0.06 €/k$Wh_{Bio}$*

# Takeaways

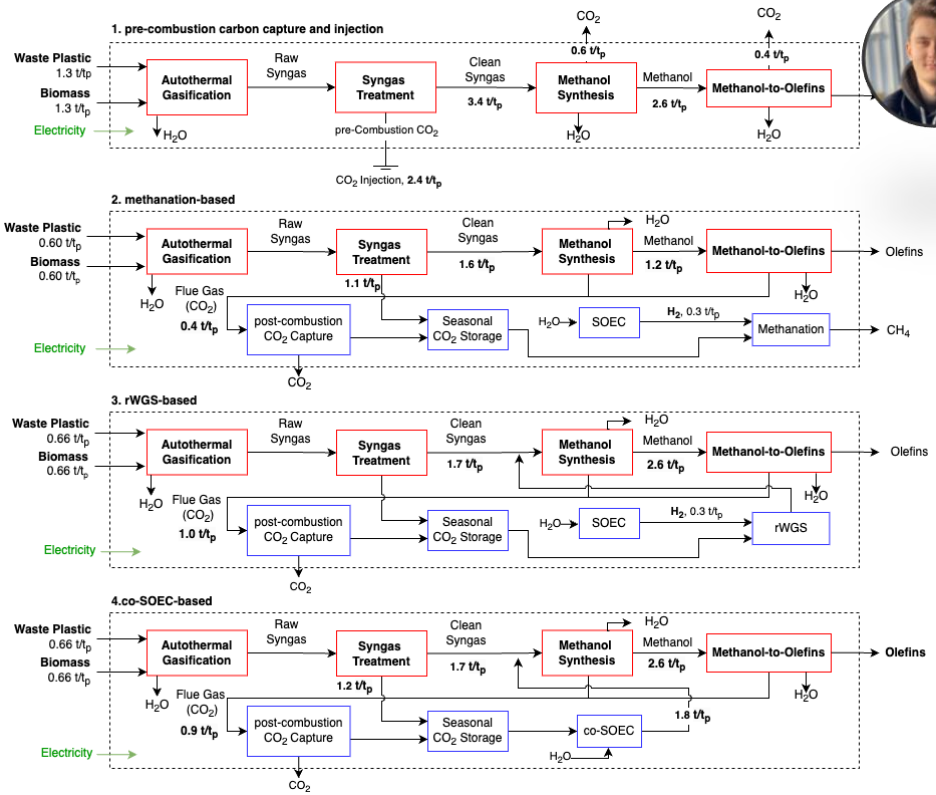
1. Plasma: Highest cost, lowest TRL.
2. Hydrogen: Indirect electrification of reverberatory furnaces.
3. Oxyfuel CCS: Lowest energy consumption and cost
4. Net-zero or negative emissions only with biomass conversion and carbon sequestration
5. Direct electrification for the rolling plant

# Chemicals sector




*Breakdown of the projected increse in plastic use by 2060 (OECD 2022)*

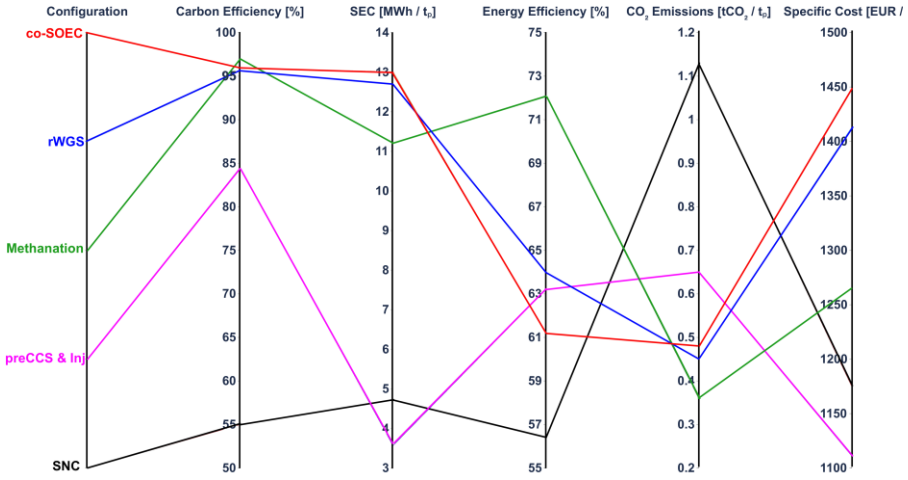
- Steam cracking of hydrocarbons predominant process for olefins production.
- 1420 Mt $CO_2$ per year is emitted ( 70% of primary chemicals*)[2]
- Waste management +0.4 kg$CO_2$e per kg of plastics recycled mechanically.




- Chemical processes were modelled in Aspen Plus® and then formulated in OSMOSE for integration of $CO_2$ abatement, energy systems, and utilities.
- Heat pump, steam network and cooling tower models are integrated for energy management.
- Renewable intermittency is considered by electricity price variation. (Nov-Feb: 0.15 EUR/kWh and Mar-Oct: 0.05 EUR/kWh

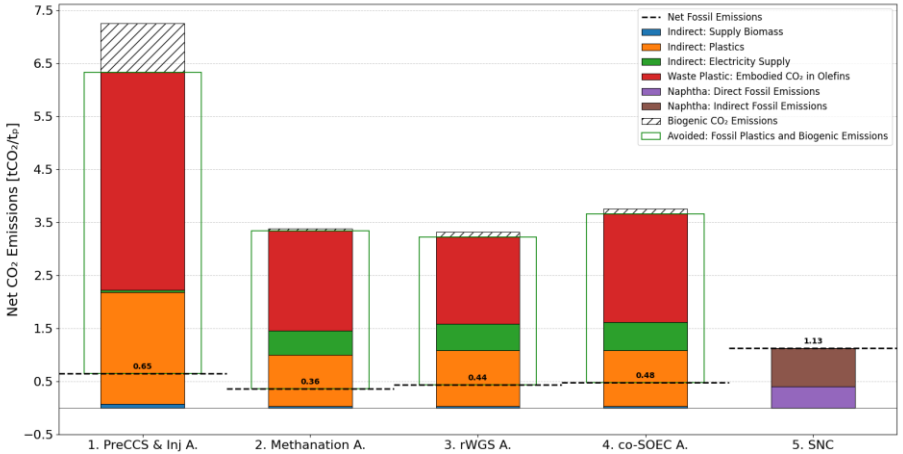
* Primary chemicals include, ammonia, plastics and methanol products

# Chemicals sector

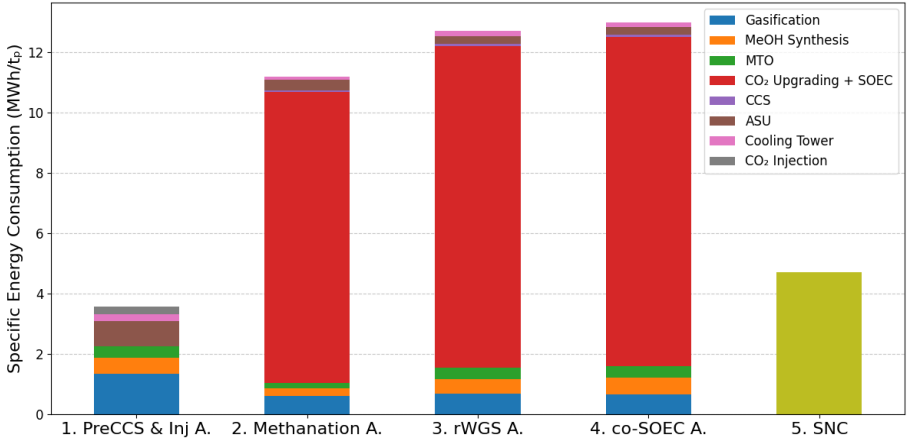



*Key performance indicators for process configurations*

- The carbon content of the feedstock is predominantly conserved within the products.
- The energy demand is fully electrified, with SOEC accounting for approximately 60–70% of the total.
- Integration of heat recovery, steam network and SOEC improves energy efficiency at least by 5%.
- Net fossil $CO_2$ emissions are reduced up to 70%.
- Plastic feedstock and electricity costs account for approximately 60–70% of the specific operating costs.




*Net $CO_2$ emissions balance for the process configurations.*

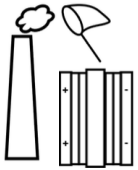



*Breakdown of the specific energy demand for the process configurations.*

# Takeaways



The circular approach maximizes carbon efficiency while preventing the introduction of virgin fossil feedstocks into the value chain.



The integration of carbon capture and utilization results in a 40–70% decrease in net fossil-based $CO_2$ emissions.



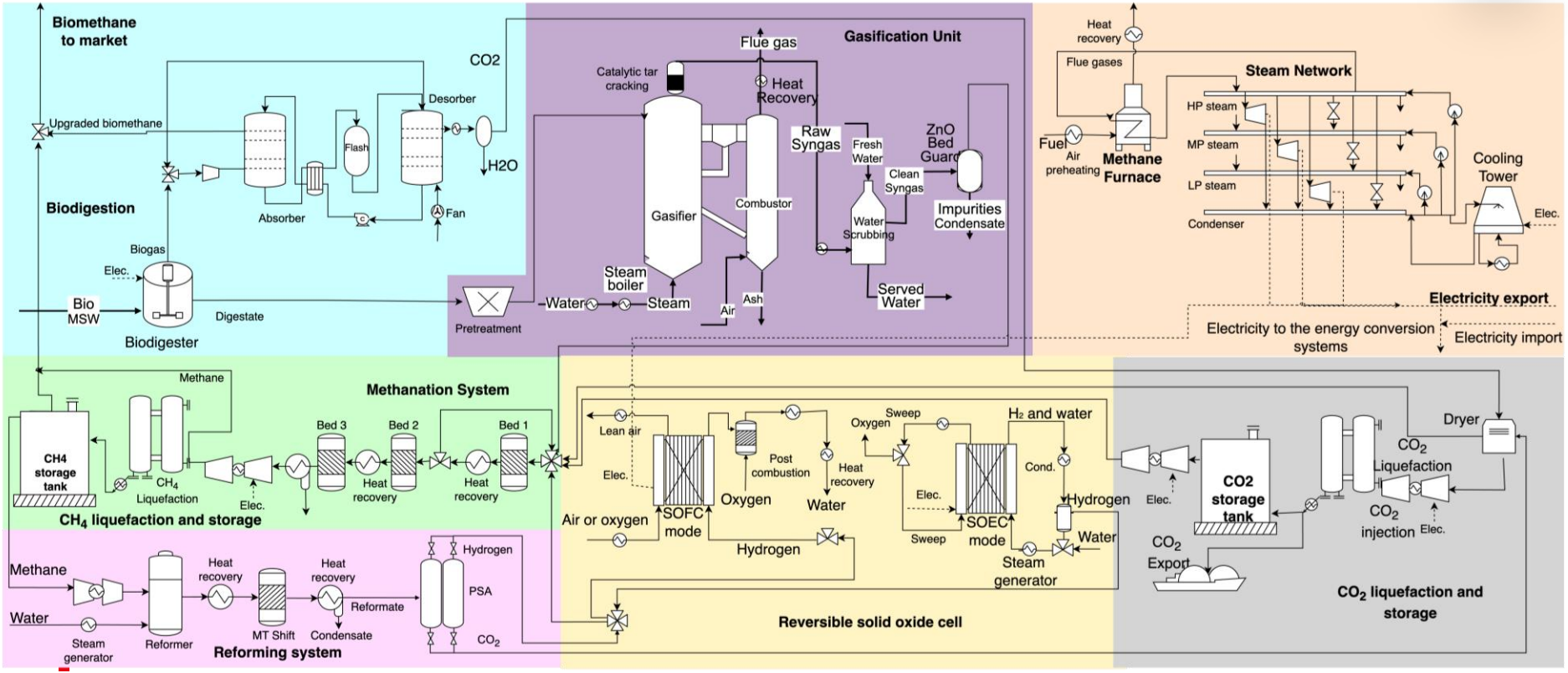
Seasonal $CO_2$ storage mitigates renewable intermittency, cutting energy costs by 27%.



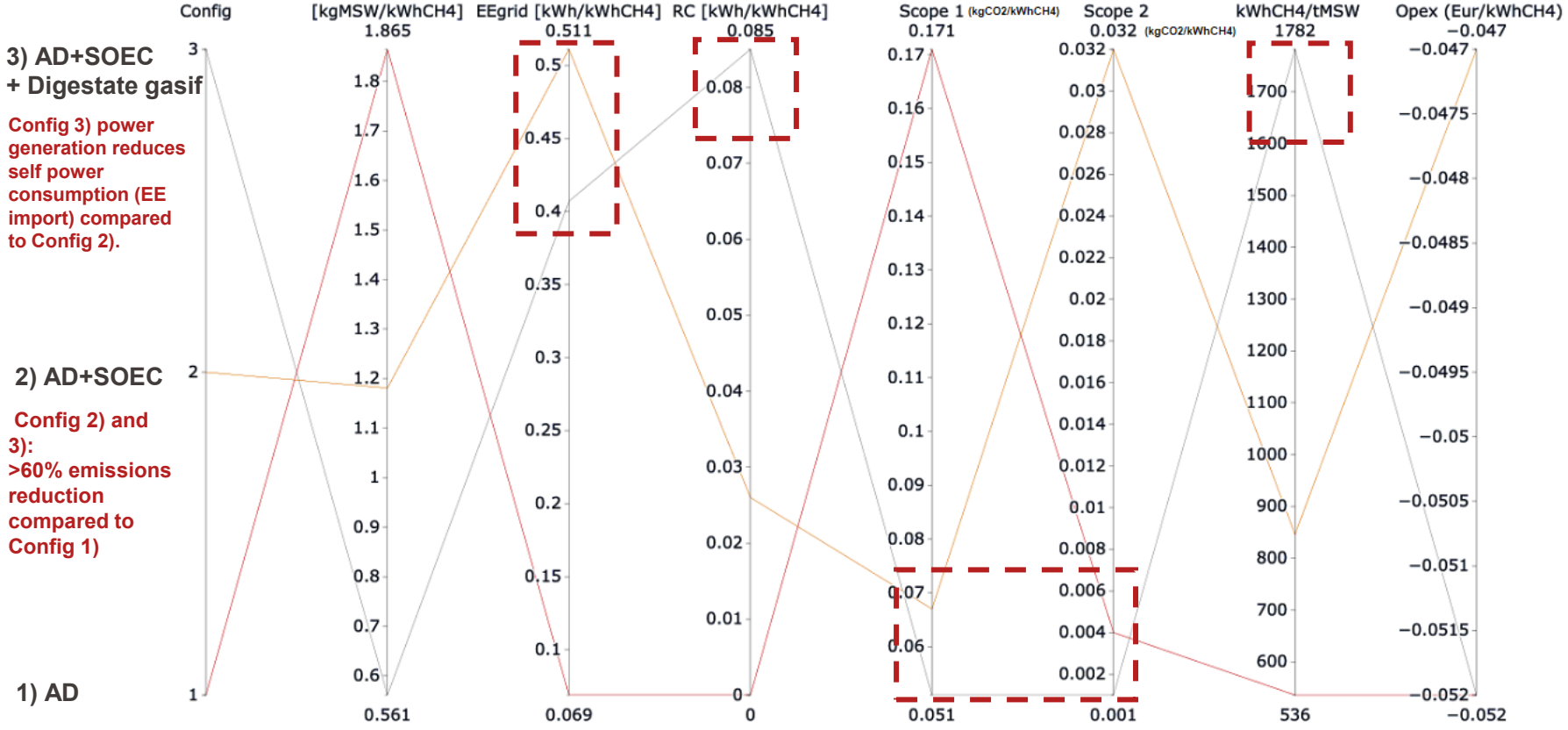
Renewable and low-cost electricity are the key drivers for the economic feasibility.

# Biowaste valorization sector





# Biowaste valorization sector




Config 3) power generation reduces self power consumption (EE import) compared to Config 2).

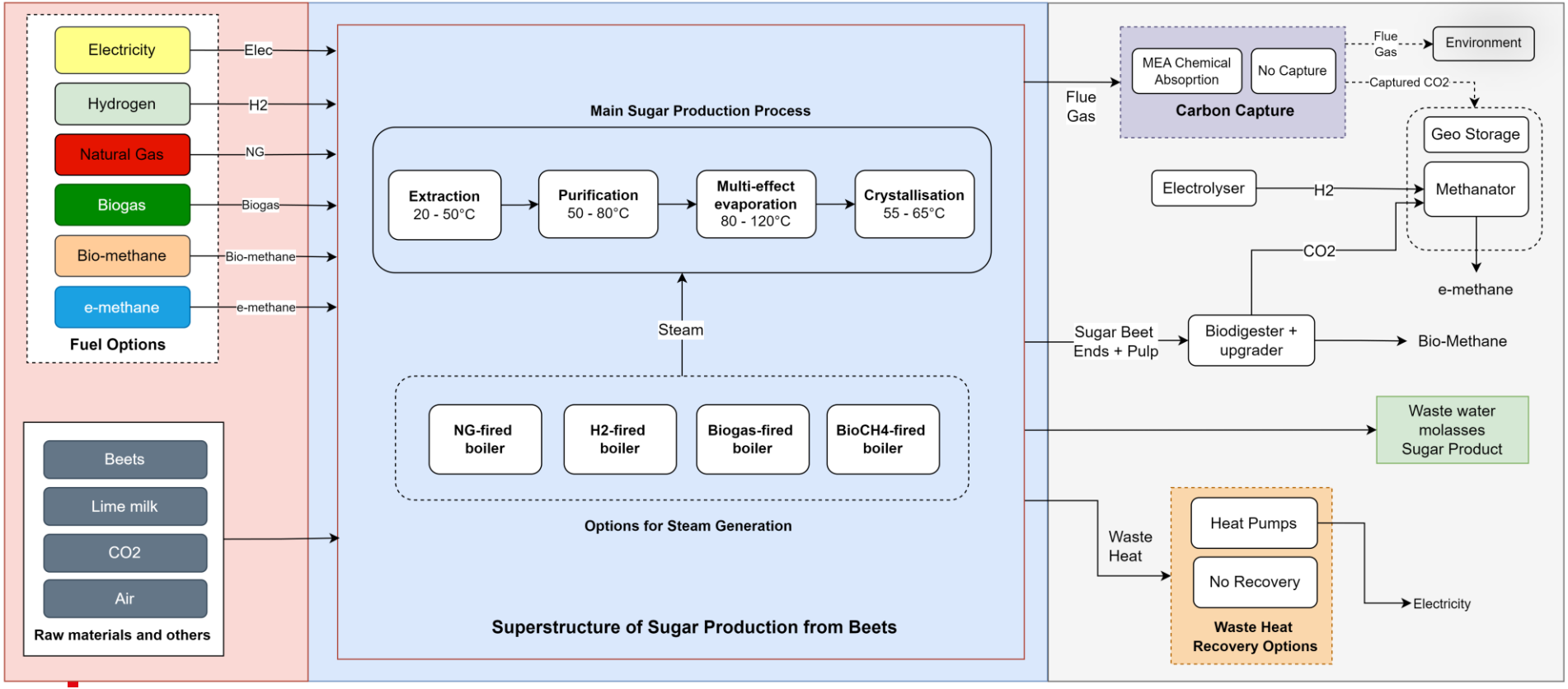
Config 2) and 3):
>60% emissions reduction compared to Config 1)

# Takeaways

- Systematic approach combining different energy solutions and time-varying energy demands to identify the operating conditions and arrangements for sustainable natural gas production.
- Power-to-gas: $CO_2$ streams are stored and utilized to produce $CH_4$ when electricity price is more attractive.
- The $CO_2$ storage enables harvesting excess renewable electricity and producing value added fuels from biogenic emissions.
- Electricity imported is used in the plant to balance the power generation of the Rankine cycle, as electricity is imported (summer period) and stored (power-to-gas).
- Self power generation is crucial for enhancing the process efficiency and waste heat recovery.

# Sugar sector

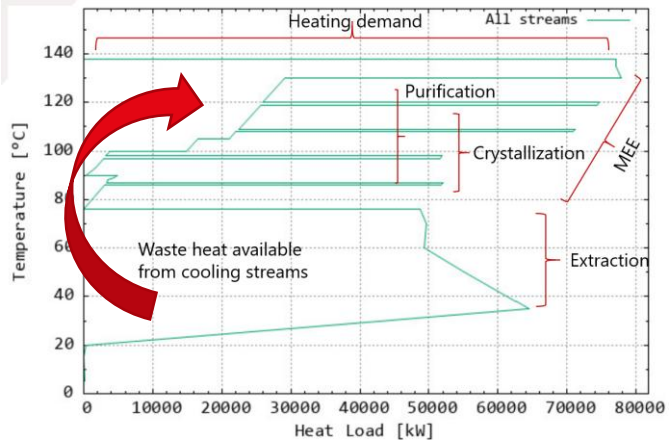


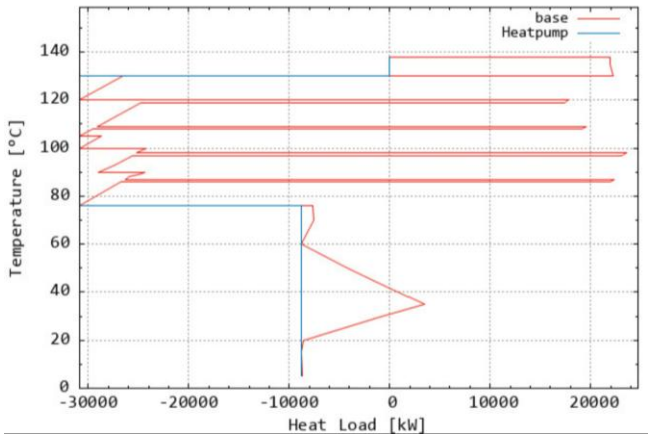


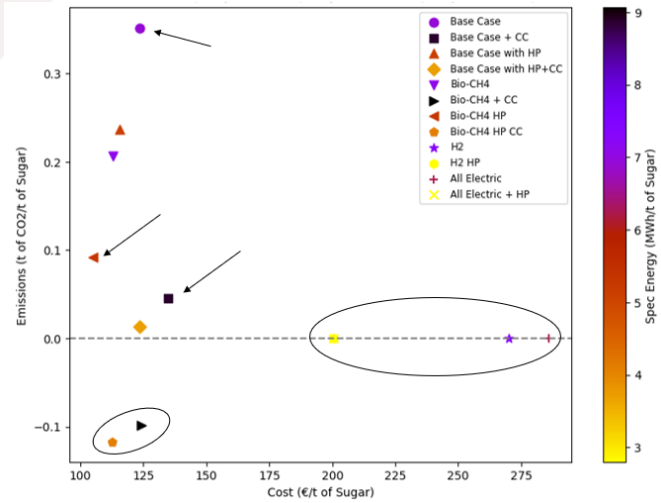
# Sugar sector

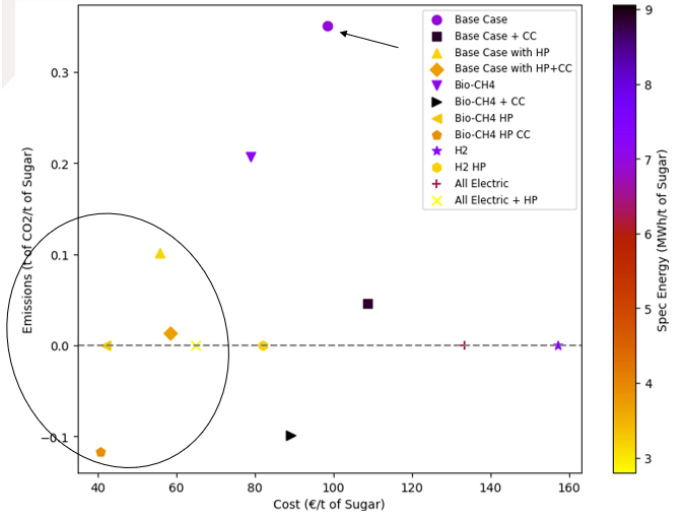
- Size of the plant = 1400 tpd of sugar
- Sugar beet ends and Pulp waste used as Biomass
- Carbon capture = MEA Chemical absorption (90% capture rate)
- Emissions = Direct + Indirect
- Capture cost = 80 €/tonne of $CO_2$*
- Scope 1 emissions do not include biogenic $CO_2$
- Objective functions
  - Minimum Total Annual Cost (€/$t_{glass}$)
  - Minimum Emissions ($t_{CO2}/t_{glass}$)

| | |
|---|---|
| No Renewable Energy Scenario **2025** | • $CO_2$ tax = 80 €/t$CO_2$<br>• Electricity Price = 100 €/MWh<br>• NG Price = 35 €/MWh |
| High Renewable Energy Scenario **2050** | • $CO_2$ tax = 250 €/t$CO_2$<br>• Electricity Price = 50 €/MWh<br>• NG Price = 35 €/MWh |



# Takeaways

## ⚡ Key Findings

- **Heat pumps** (HPs) significantly **reduce energy use and emissions**.
- **Bio-$CH_4$ + HPs** is the **most cost-effective and low-emission** route (net-negative possible with CC).
- **Electric boilers** become competitive only with HP integration and **cheap electricity**.
- **Hydrogen is not viable** due to high cost and infrastructure needs.

## 💡 Role of Electricity Prices

- As electricity prices drop (2030, 2050 scenarios):
  - Electrification becomes economically viable
  - **HP-based routes outperform** NG and hydrogen
- If very low-cost electricity is available:
  - **Captured $CO_2$** from on-site bio-$CH_4$ + CC **can be stored and reused** to synthesize e-$CH_4$
  - Enables **closed-loop**, fossil-free energy cycle

## ✅ Conclusions

- **HPs are essential** for energy-efficient decarbonization
- **Bio-$CH_4$ enables carbon-negative** sugar production
- **Variable electricity pricing unlocks smart electrification**
- Superstructure optimization helps balance **trade-offs between cost, emissions, and energy**

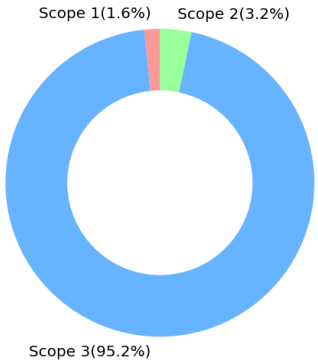
# Watch industry



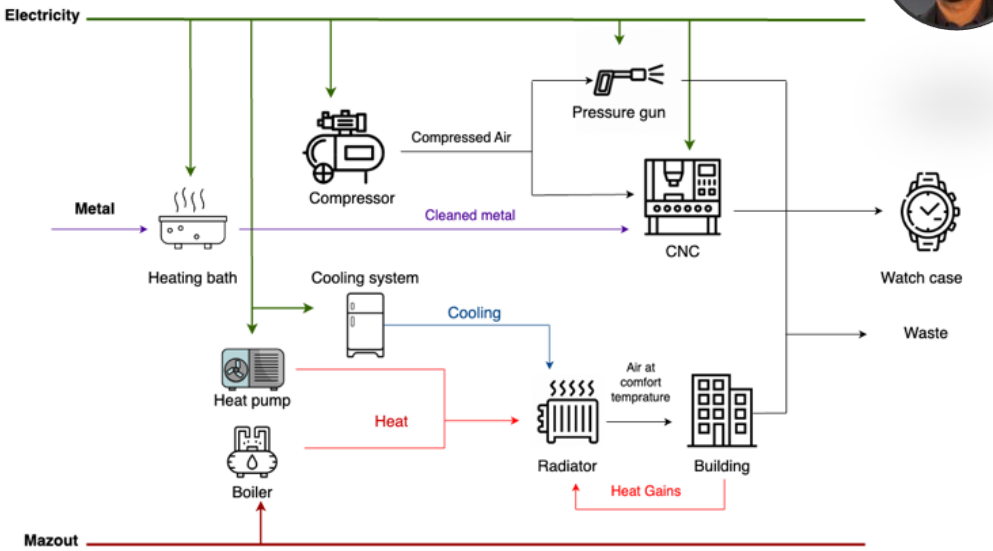


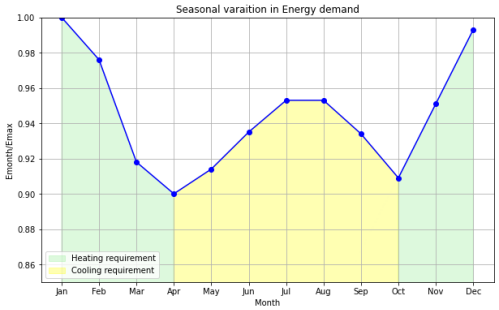


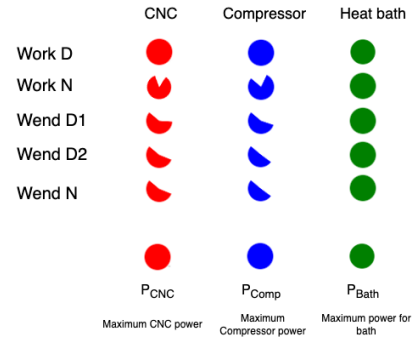
1 billion watches




| Background processes | Watch 1 | Watch 2 |
|---|---|---|
| Case | Stainless steel | Stainless Steel, Gold, Platinum |
| Strap | Stainless steel | Stainless Steel, Gold, Platinum |
| Covering | Glass fibre | Glass fibre |



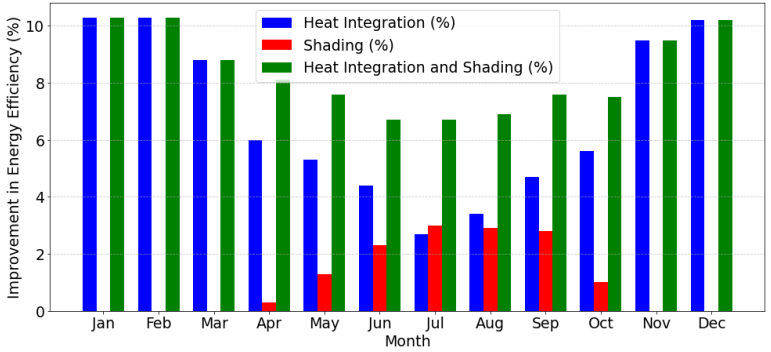




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Reference: Swatch group Sustaianability report 2024

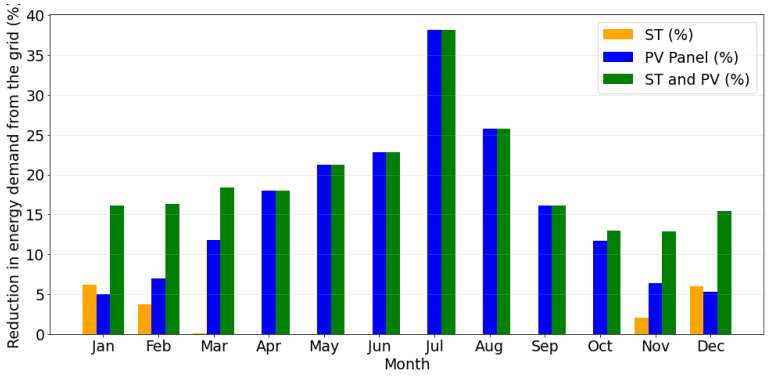
# Watch industry

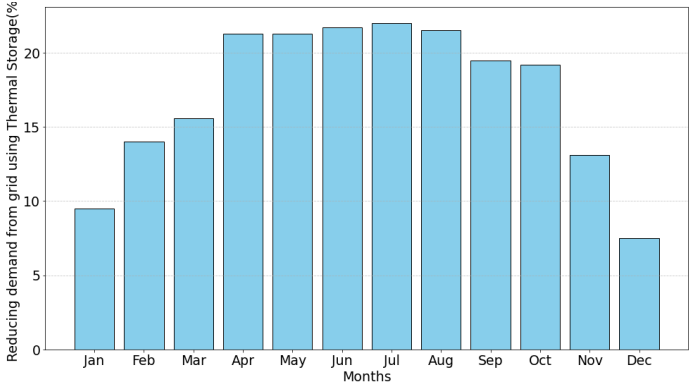
## Improving heat recovery and seasonality




| Technology | CAPEX (kCHF/year) | OPEX sav. (kCHF/year) |
|---|---|---|
| BAU | 0 | 0 |
| PV | 122 | -219 |
| ST | 122 | -109 |
| PVT | 229 | -318 |
| Thermal storage | 122 | -166 |

## Integrating Decarbonization option and seasonality

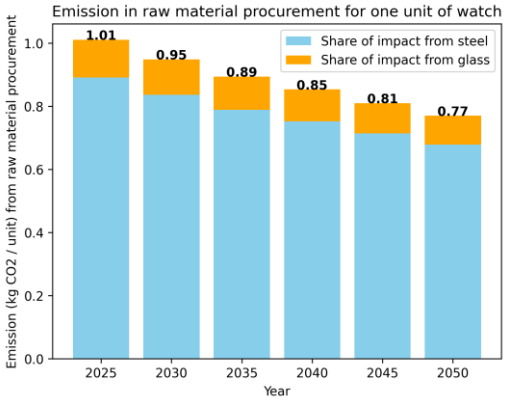


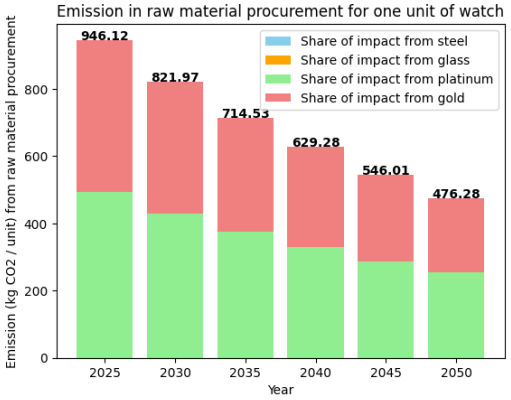




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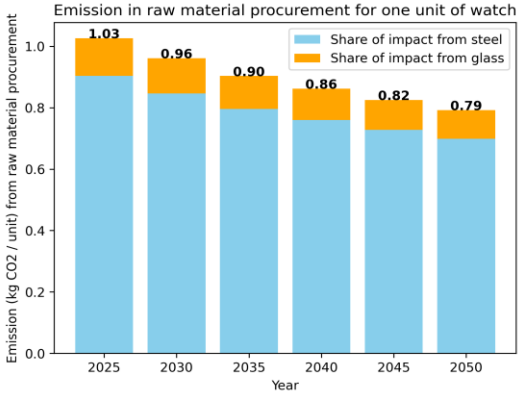
# Watch production background

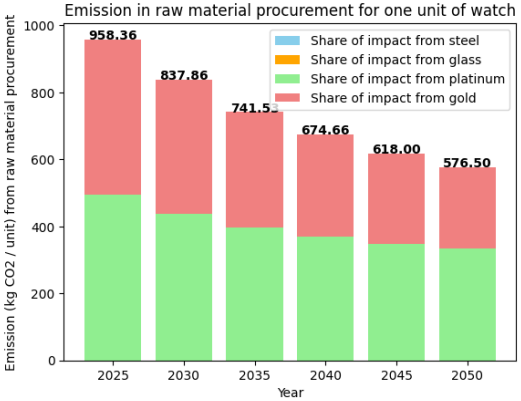
## SSP-1



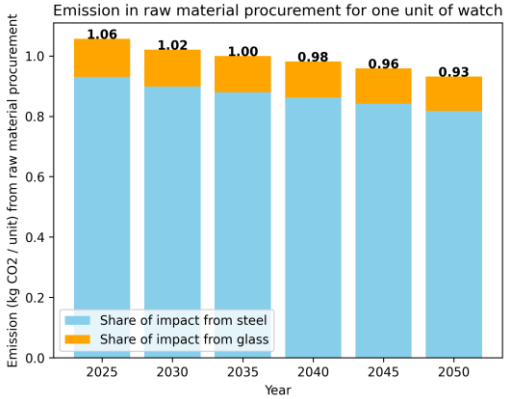



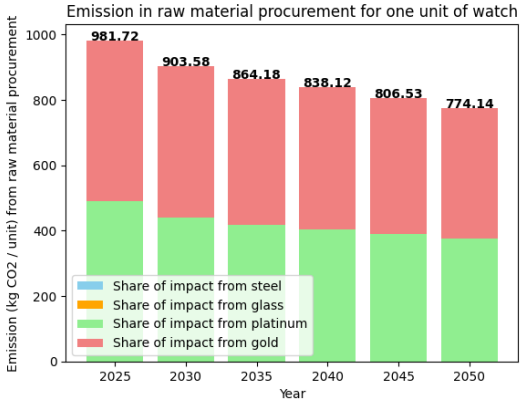

## SSP-2




## SSP-5




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# Conclusions

- Challenge: sharing not merely data, but also models and tools for existing and new processes and technologies relevant for the energy transition.
- Procedures for sharing models and energy integration routines to generate, evaluate, hierarchize and document multiple and attractive system configurations.
- Validated ex-ante models allow to determine the opportunities for the industrial symbiosis between production processes, waste management and decarbonization technologies.
- Define sustainable performance metrics for decision support, ensuring objective and reproducible assessment and reports.
- Benefits: (i) centralized library of **validated** models; (ii) easy **maintenance** and upgrade; (iii) constant improvement by **sharing** knowledge; and (iv) facilitated browsing and **comparison** between technological solutions.

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